

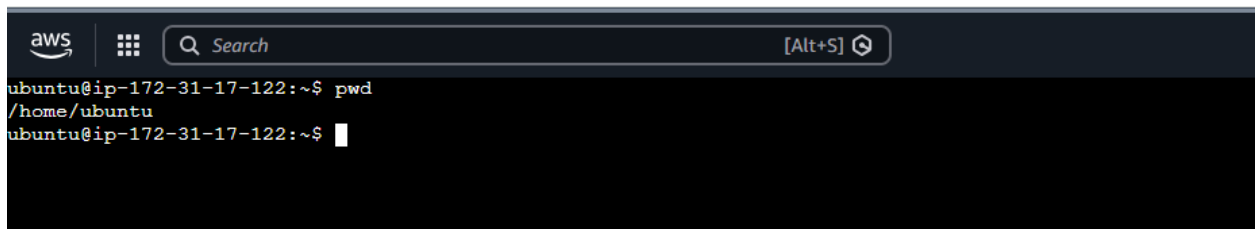
GitHub link : [http://github.com/SujataHoonalli/MY\\_GITHUB\\_WORK](http://github.com/SujataHoonalli/MY_GITHUB_WORK)

## 1. L1 - In EC2 Ubuntu Instance Create a new user and SSH Key pair with an authorized key

Step 1: Connect to AWS EC2 Ubuntu instance.

Step 2: Clear the terminal

Run command: clear



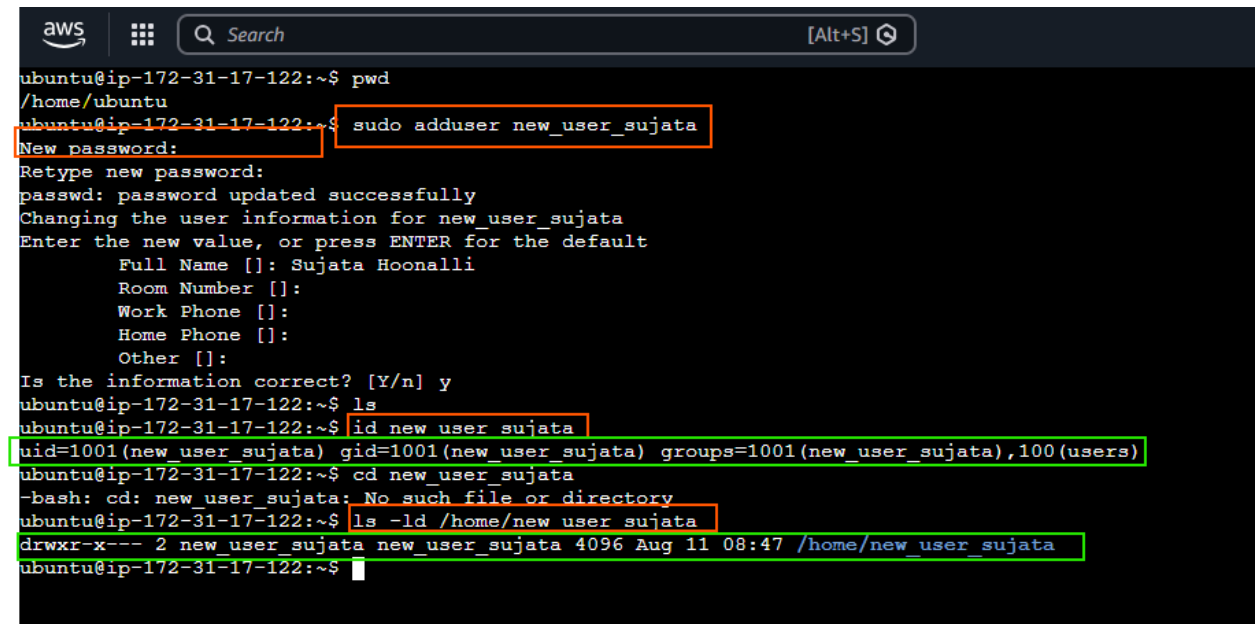
```
aws | Search [Alt+S]
ubuntu@ip-172-31-17-122:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-17-122:~$
```

the default **ubuntu** user

Step 3: Create new user

Run command : `sudo adduser <new_user_name>`

after this it will ask for password and new value details as shown in below image



```
aws | Search [Alt+S]
ubuntu@ip-172-31-17-122:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-17-122:~$ sudo adduser new_user_sujata
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for new_user_sujata
Enter the new value, or press ENTER for the default
  Full Name []: Sujata Hoonalli
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
ubuntu@ip-172-31-17-122:~$ ls
ubuntu@ip-172-31-17-122:~$ id new user sujata
uid=1001(new_user_sujata) gid=1001(new_user_sujata) groups=1001(new_user_sujata),100(users)
ubuntu@ip-172-31-17-122:~$ cd new_user_sujata
-bash: cd: new_user_sujata: No such file or directory
ubuntu@ip-172-31-17-122:~$ ls -ld /home/new user sujata
drwxr-x--- 2 new_user_sujata new_user_sujata 4096 Aug 11 08:47 /home/new_user_sujata
ubuntu@ip-172-31-17-122:~$
```

To check whether new user is created or not , Run : `id <new_user_name>`

Or `ls -ld /home/<new_user_name>`

If we get expected result as highlighted in green , it means new user is created properly

Step 4: Now switch to new user

Run command : su <new\_user\_name>

It will ask for password, enter the password which is already set in “Step 3”

```
ubuntu@ip-172-31-17-122:~$ su new_user_sujata
Password:
new user sujata@ip-172-31-17-122:/home/ubuntu$ ls
```

Now observe user is changed from **ubutu** to **new\_user\_sujata**

Step 5: Now generate ssh key pair

Run command : ssh-keygen

It will start generating public and private key

It will ask for file creation path and save path and passphrase.

Here I set it as default .

```
new_user_sujata@ip-172-31-17-122:/$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/new_user_sujata/.ssh/id_ed25519):
Created directory '/home/new_user_sujata/.ssh'.
Enter passphrase for "/home/new_user_sujata/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/new_user_sujata/.ssh/id_ed25519
Your public key has been saved in /home/new_user_sujata/.ssh/id_ed25519.pub
The key's fingerprint is:
SHA256:cGbVtVWzss6JjKCQnpEX0i5MO8zDsa30yqIJzd98PQ0 new_user_sujata@ip-172-31-17-122
The key's randomart image is:
+--[ED25519 256]--+
|      .+..oo |
|      .. o o |
|      .+   o |
|      o  =E  o |
|      = + .So . |
|      @ O ..o... |
|      |. ^ +.o o.+ |
|      |oo.X.o o . |
|      |to=.o... |
+-----[SHA256]-----+
new_user_sujata@ip-172-31-17-122:/$ cd .ssh
bash: cd: .ssh: No such file or directory
new_user_sujata@ip-172-31-17-122:/$ ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin snap srv sys usr var
new_user_sujata@ip-172-31-17-122:/$ cd home/new_user_sujata/
new_user_sujata@ip-172-31-17-122:~$ ls
new_user_sujata@ip-172-31-17-122:~$ ll
total 24
drwxr-x--- 3 new_user_sujata new_user_sujata 4096 Aug 11 09:07 ./
drwxr-xr-x 4 root root 4096 Aug 11 08:47 ../
-rw-r--r-- 1 new_user_sujata new_user_sujata 220 Aug 11 08:47 .bash_logout
-rw-r--r-- 1 new_user_sujata new_user_sujata 3771 Aug 11 08:47 .bashrc
-rw-r--r-- 1 new_user_sujata new_user_sujata 807 Aug 11 08:47 .profile
drwx----- 2 new_user_sujata new_user_sujata 4096 Aug 11 09:07 .ssh/
new_user_sujata@ip-172-31-17-122:~$ cd .ssh
new_user_sujata@ip-172-31-17-122:~/.ssh$ ll
total 16
drwx----- 2 new_user_sujata new_user_sujata 4096 Aug 11 09:07 ./
drwxr-x--- 3 new_user_sujata new_user_sujata 4096 Aug 11 09:07 ../
-rw----- 1 new_user_sujata new_user_sujata 432 Aug 11 09:07 id_ed25519
-rw-r--r-- 1 new_user_sujata new_user_sujata 114 Aug 11 09:07 id_ed25519.pub
new_user_sujata@ip-172-31-17-122:~/.ssh$
```

Once key is generated, go to particular path, here cd home/new\_user\_sujata

We can see .ssh folder is created and inside that private and public key pair also generated.

Step 6: Create the authorized\_keys file by copying the public key:

Run the command: cp id\_ed25519.pub authorized\_keys

If the output matches id\_ed25519.pub, then the public key has been authorized

```
new_user_sujata@ip-172-31-17-122:~/.ssh$ cp id_ed25519.pub authorized_keys
new_user_sujata@ip-172-31-17-122:~/.ssh$ ls
authorized_keys  id_ed25519  id_ed25519.pub
new_user_sujata@ip-172-31-17-122:~/.ssh$ cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIEXWER4RrQrcghMeM8U/dbKie8cD8vcC+3xwDMHVwdEK new_user_sujata@ip-172-31-17-122
new_user_sujata@ip-172-31-17-122:~/.ssh$ ll
total 20
drwx----- 2 new_user_sujata new_user_sujata 4096 Aug 11 09:19 ./
drwxr-x--- 3 new_user_sujata new_user_sujata 4096 Aug 11 09:07 ../
-rw-r--r-- 1 new_user_sujata new_user_sujata 114 Aug 11 09:19 authorized_keys
-rw----- 1 new_user_sujata new_user_sujata 432 Aug 11 09:07 id_ed25519
-rw-r--r-- 1 new_user_sujata new_user_sujata 114 Aug 11 09:07 id_ed25519.pub
new_user_sujata@ip-172-31-17-122:~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIEXWER4RrQrcghMeM8U/dbKie8cD8vcC+3xwDMHVwdEK new_user_sujata@ip-172-31-17-122
new_user_sujata@ip-172-31-17-122:~/.ssh$
```

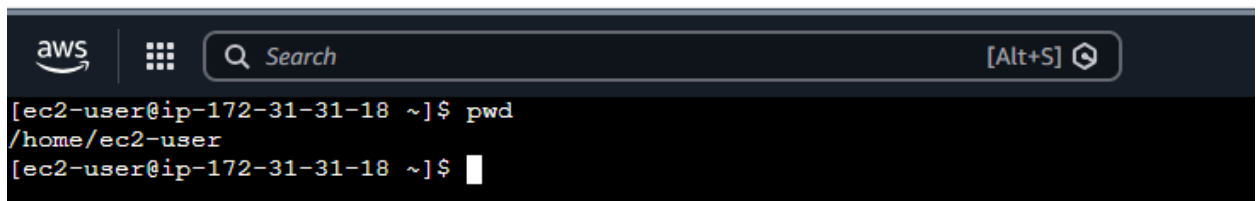
|===== L1 completed =====|

## 2. L2 - As a Linux root user Create Files/Directory in the same Instance and change the ownership to a new user

Step 1: Connect to AWS EC2 linux instance.

Step 2: Clear the terminal

Run command: clear



```
aws | Search [Alt+S]
[ec2-user@ip-172-31-31-18 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-31-18 ~]$
```

Step 3: Switch to root directory from home directory

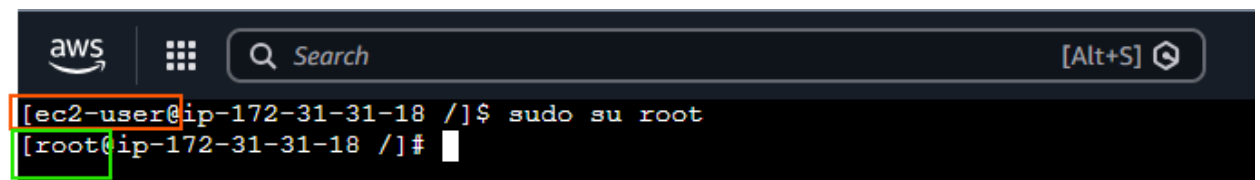
Run command: cd /



```
aws | Search [Alt+S]
[ec2-user@ip-172-31-31-18 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-31-18 ~]$ cd /
[ec2-user@ip-172-31-31-18 /]$ ls
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys tmp usr var
[ec2-user@ip-172-31-31-18 /]$
```

Step 4: Become a root user

Run command: sudo su root



```
aws | Search [Alt+S]
[ec2-user@ip-172-31-31-18 /]$ sudo su root
[root@ip-172-31-31-18 /]#
```

Step 5: Create a main folder and files inside that

Run command : mkdir <folder\_name> to create directory

Run command : touch <file\_name1> <file\_name2> to create files inside the main folder.

```
[root@ip-172-31-31-18 /]# mkdir main_folder
[root@ip-172-31-31-18 /]# cd main_folder
[root@ip-172-31-31-18 main_folder]# ls
[root@ip-172-31-31-18 main_folder]# touch file1.sh file2.sh
[root@ip-172-31-31-18 main_folder]# ls
file1.sh  file2.sh
[root@ip-172-31-31-18 main_folder]#
```

```
[root@ip-172-31-31-18 /]# cd main_folder/
[root@ip-172-31-31-18 main_folder]# ll
total 0
-rw-r--r--. 1 root root 0 Aug 11 10:29 file1.sh
-rw-r--r--. 1 root root 0 Aug 11 10:29 file2.sh
[root@ip-172-31-31-18 main_folder]#
```

#### Step 6: change the ownership to a new user

Run command : ls /home

If there is no new user present,

Create a new user by running: sudo adduser <user\_name>

```
[root@ip-172-31-31-18 /]# mkdir main_folder
[root@ip-172-31-31-18 /]# cd main_folder
[root@ip-172-31-31-18 main_folder]# ls
[root@ip-172-31-31-18 main_folder]# touch file1.sh file2.sh
[root@ip-172-31-31-18 main_folder]# ls
file1.sh  file2.sh
[root@ip-172-31-31-18 main_folder]# ls /home
ec2-user
[root@ip-172-31-31-18 main_folder]# sudo adduser new_user_sujata
[root@ip-172-31-31-18 main_folder]# ls /home
ec2-user  new_user_sujata
[root@ip-172-31-31-18 main_folder]#
```

**Before :**

```
[root@ip-172-31-31-18 /]# cd main_folder/
[root@ip-172-31-31-18 main_folder]# ll
total 0
-rw-r--r--. 1 root root 0 Aug 11 10:29 file1.sh
-rw-r--r--. 1 root root 0 Aug 11 10:29 file2.sh
[root@ip-172-31-31-18 main_folder]#
```

**Run :** `chown -R new_user_sujata:new_user_sujata /main_folder`

To change owner access from **root** to **new\_user\_sujata**

**After :**

```
[root@ip-172-31-31-18 main_folder]# chown -R new_user_sujata:new_user_sujata /main_folder
[root@ip-172-31-31-18 main_folder]# ls -ll /main_folder
total 0
-rw-r--r--. 1 new_user_sujata new_user_sujata 0 Aug 11 10:29 file1.sh
-rw-r--r--. 1 new_user_sujata new_user_sujata 0 Aug 11 10:29 file2.sh
[root@ip-172-31-31-18 main_folder]#
```

|===== L2 completed =====|

### 3. L3 - In EC2 Ubuntu Instance Create Files and Directories and Grand R/W/X Access only to the Owner and User Group

Step 1: Connect to AWS EC2 Ubuntu instance.

Step 2: Clear the terminal

Run command: clear

Step 3: Create directories and file

```
ubuntu@ip-172-31-23-18:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-23-18:~$ ll
total 32
drwxr-x--- 4 ubuntu ubuntu 4096 Aug 13 11:52 ./
drwxr-xr-x 3 root    root   4096 Aug 13 11:47 ../
-rw----- 1 ubuntu ubuntu   98 Aug 13 11:51 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .ssh/
ubuntu@ip-172-31-23-18:~$ mkdir folder1 folder2
ubuntu@ip-172-31-23-18:~$ touch file1.sh file2.sh
ubuntu@ip-172-31-23-18:~$ ll
total 40
drwxr-x--- 6 ubuntu ubuntu 4096 Aug 13 11:53 ./
drwxr-xr-x 3 root    root   4096 Aug 13 11:47 ../
-rw----- 1 ubuntu ubuntu   98 Aug 13 11:51 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .ssh/
-rw-rw-r-- 1 ubuntu ubuntu    0 Aug 13 11:53 file1.sh
-rw-rw-r-- 1 ubuntu ubuntu    0 Aug 13 11:53 file2.sh
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 13 11:52 folder1/
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 13 11:52 folder2/
ubuntu@ip-172-31-23-18:~$
```

Step 4: Give r/w/x only to the Owner and User Group

Run command : `chmod 770 folder1 folder2` (for directories)

Run command : `chmod 770 file1.sh file2.sh` (for files)

```
ubuntu@ip-172-31-23-18:~$ chmod 770 folder1 folder2
ubuntu@ip-172-31-23-18:~$ ll
total 40
drwxr-x--- 6 ubuntu ubuntu 4096 Aug 13 11:53 ./
drwxr-xr-x 3 root    root   4096 Aug 13 11:47 ../
-rw----- 1 ubuntu ubuntu   98 Aug 13 11:51 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .ssh/
-rw-rw-r-- 1 ubuntu ubuntu    0 Aug 13 11:53 file1.sh
-rw-rw-r-- 1 ubuntu ubuntu    0 Aug 13 11:53 file2.sh
drwxrwx--- 2 ubuntu ubuntu 4096 Aug 13 11:52 folder1/
drwxrwx--- 2 ubuntu ubuntu 4096 Aug 13 11:52 folder2/
ubuntu@ip-172-31-23-18:~$ chmod 770 file1.sh file2.sh
ubuntu@ip-172-31-23-18:~$ ll
total 40
drwxr-x--- 6 ubuntu ubuntu 4096 Aug 13 11:53 ./
drwxr-xr-x 3 root    root   4096 Aug 13 11:47 ../
-rw----- 1 ubuntu ubuntu   98 Aug 13 11:51 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Aug 13 11:47 .ssh/
-rwxrwx--- 1 ubuntu ubuntu    0 Aug 13 11:53 file1.sh*
-rwxrwx--- 1 ubuntu ubuntu    0 Aug 13 11:53 file2.sh*
drwxrwx--- 2 ubuntu ubuntu 4096 Aug 13 11:52 folder1/
drwxrwx--- 2 ubuntu ubuntu 4096 Aug 13 11:52 folder2/
ubuntu@ip-172-31-23-18:~$
```

|===== L3 completed =====|

## L4 - In EC2 Ubuntu Instance install JDK and setup JAVA\_HOME path environment variable

Step 1: Connect to AWS EC2 Ubuntu instance.

Step 2: Clear the terminal

Run command: clear

Step 3 : Update Ubuntu packages

Run command : sudo apt update -y

Step 4: Install jdk

Run command : sudo apt install openjdk-25-jre-headless -y

Step 5 : verify installation

Run command: java -version

Step 6 :

**Find the path, where java is installed and executable**

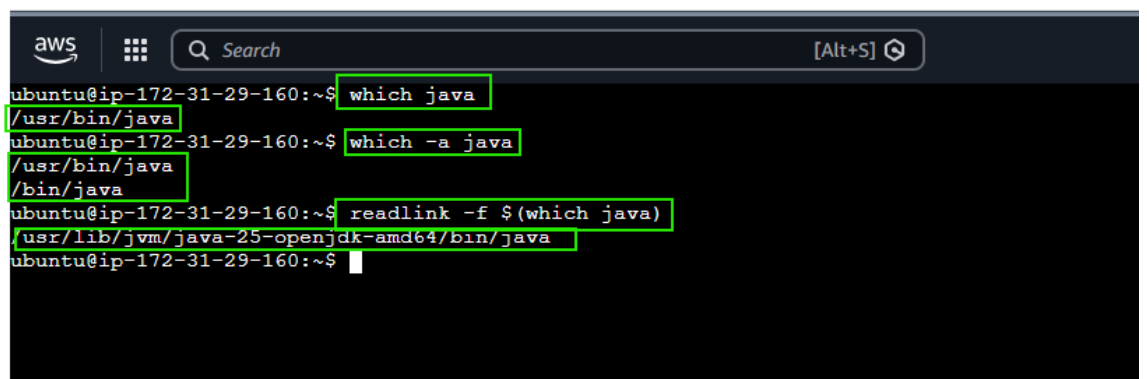
Run command : which java

**To see all the java executables in our PATH**

Run command : which -a java

**To find actual java installation directory**

Run command : readlink -f \$(which java)



```
aws | Search [Alt+S]
ubuntu@ip-172-31-29-160:~$ which java
/usr/bin/java
ubuntu@ip-172-31-29-160:~$ which -a java
/usr/bin/java
/bin/java
ubuntu@ip-172-31-29-160:~$ readlink -f $(which java)
/usr/lib/jvm/java-25-openjdk-amd64/bin/java
ubuntu@ip-172-31-29-160:~$
```

Step 6: Check echo \$JAVA\_HOME

If its empty, add **setup JAVA\_HOME path environment variable**

By running export JAVA\_HOME = /usr/lib/jvm/java-25-openjdk-amd64

Then run : echo PATH=\$JAVA\_HOME/bin:\$PATH

Then Run : echo \$PATH, To get path

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&iinstanceId=i-0e7b1f60b9afa125c&osUser=ubuntu&sshPort=22&addressFami

aws [Alt+S]
ubuntu@ip-172-31-29-160:~$ which java
/usr/bin/java
ubuntu@ip-172-31-29-160:~$ which -a java
/usr/bin/java
/bin/java
ubuntu@ip-172-31-29-160:~$ readlink -f $(which java)
/usr/lib/jvm/java-25-openjdk-amd64/bin/java
ubuntu@ip-172-31-29-160:~$ echo $JAVA_HOME
ubuntu@ip-172-31-29-160:~$ export JAVA_HOME=/usr/lib/jvm/java-25-openjdk-amd64
ubuntu@ip-172-31-29-160:~$ echo PATH=$JAVA_HOME/bin:$PATH
PATH=/usr/lib/jvm/java-25-openjdk-amd64/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
ubuntu@ip-172-31-29-160:~$ echo $JAVA_HOME
/usr/lib/jvm/java-25-openjdk-amd64
ubuntu@ip-172-31-29-160:~$
```

===== L4 completed =====

## L5 - Create two AWS EC2 Ubuntu Instances to establish SSH Connection and SCP the files from one Instance to another instance

Step 1: Create 2 AWS EC2 Ubuntu Instances

In Ubuntu instance 1 :

Step 2: Create key pair

Run command: `ssh-keygen -t rsa -b 4096`

Step 3 : copy .pub key into authorized\_keys

```
ubuntu@ip-172-31-17-136:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase for '/home/ubuntu/.ssh/id_rsa' (empty for no passphrase):
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
60:25:41:22:05:46:00:13:03:0C:3B:00:32:FA:5B:DD:8B:5F:46:4A:0C:3D:00:6B:3F:W2AJA:5f3KX:lyG1:GfYogA+IHome3
The key's randomart image is:
+----[RSA 4096]-----+
|  ..+..          |
|  +..+..         |
|  o o o . o .    |
|  + o . . .      |
|  = + o .        |
|  o + =ooo o     |
|  . + .+o+       |
|  .+o .o+        |
|  .+o .o+o+      |
|  +-----[SHA256]-----+
ubuntu@ip-172-31-17-136:~$ cd .ssh/
ubuntu@ip-172-31-17-136:~/.ssh$ cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDgPTFg7DdRxtaklJWbhc+25lW/IB+wa743Vog2hOxwE6b6fkxiasPtnlc0uX0RPF85Y8ma2EacHUFDVDR0/opnxttkcEB4w0zmaKv8vz+C/N7L5HK+xaIVcVqg8FWbHtSKH6mjlV425amNA6FFyzsMLI3VChnUQ/tkTm1RB1
2XZT9DQqg7Gw79m4wFp8KXh5lUfakgWw+28V3Ww1r0u2lU0DFGfNf6+YI:Ufj13Kgz+TK84i60B3D1l0w7cER2djkqPwO/1349Dq+vmuffwQ5tFPG24CkqWwYk87GzshlKxf7NLAf58HXh8hF14QLA0Ck3QDwob3FW2AJA:5f3KX:lyG1:GfYogA+IHome3
Fv2jyVQwKd8xK3DgQ82Inkq8h8hIKfJGfJgCtL0lFvAlDu6Gn54vVaMa89k84fTAZTHDHPwXkTm5SgBn5cNL02ONWKK3W81Rge48wWagj0Kc2w7zt5b6tndPrk7a9C05+Wnj51be78D82hVJXWBE01Qhrw6reaNW/sG8R0wiftrBcbvutPjyFmSpAkng+FW0zHDJ
ZRL1H0Wgbytt7T8GrcqWgVWg8Cv8Nfg7lF72G10G02c0u8HtXy158G0apPhQcpjwGphVndLcikiDptdrfCw== ubuntu@ip-172-31-17-136
ubuntu@ip-172-31-17-136:~/.ssh$ cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDgPTFg7DdRxtaklJWbhc+25lW/IB+wa743Vog2hOxwE6b6fkxiasPtnlc0uX0RPF85Y8ma2EacHUFDVDR0/opnxttkcEB4w0zmaKv8vz+C/N7L5HK+xaIVcVqg8FWbHtSKH6mjlV425amNA6FFyzsMLI3VChnUQ/tkTm1RB1
2XZT9DQqg7Gw79m4wFp8KXh5lUfakgWw+28V3Ww1r0u2lU0DFGfNf6+YI:Ufj13Kgz+TK84i60B3D1l0w7cER2djkqPwO/1349Dq+vmuffwQ5tFPG24CkqWwYk87GzshlKxf7NLAf58HXh8hF14QLA0Ck3QDwob3FW2AJA:5f3KX:lyG1:GfYogA+IHome3
Fv2jyVQwKd8xK3DgQ82Inkq8h8hIKfJGfJgCtL0lFvAlDu6Gn54vVaMa89k84fTAZTHDHPwXkTm5SgBn5cNL02ONWKK3W81Rge48wWagj0Kc2w7zt5b6tndPrk7a9C05+Wnj51be78D82hVJXWBE01Qhrw6reaNW/sG8R0wiftrBcbvutPjyFmSpAkng+FW0zHDJ
ZRL1H0Wgbytt7T8GrcqWgVWg8Cv8Nfg7lF72G10G02c0u8HtXy158G0apPhQcpjwGphVndLcikiDptdrfCw== ubuntu@ip-172-31-17-136
```

In ubuntu instance 2:

Step 4 : Add the .pub key into authorized\_keys of ubuntu instance 2

```
aws [Search] [Alt+S] United States (N. Virginia) Sujata Hoornalla (679730211815) Sujata Hoornalla
ubuntu@ip-172-31-17-184:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-17-184:~$ cd .ssh
ubuntu@ip-172-31-17-184:~/.ssh$ ll
total 24
drwx----- 2 ubuntu ubuntu 4096 Aug 14 12:42 ./
drwxr-xr-x 4 ubuntu ubuntu 4096 Aug 14 12:47 ../
-rw----- 1 ubuntu ubuntu 12288 Aug 14 12:38 authorized_keys.scp
-rw----- 1 ubuntu ubuntu 749 Aug 14 12:42 authorized_keys
ubuntu@ip-172-31-17-184:~/.ssh$ cat authorized_key
cat: authorized_key: No such file or directory
ubuntu@ip-172-31-17-184:~/.ssh$ cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDgPTFg7DdRxtaklJWbhc+25lW/IB+wa743Vog2hOxwE6b6fkxiasPtnlc0uX0RPF85Y8ma2EacHUFDVDR0/opnxttkcEB4w0zmaKv8vz+C/N7L5HK+xaIVcVqg8FWbHtSKH6mjlV425amNA6FFyzsMLI3VChnUQ/tkTm1RB1
2XZT9DQqg7Gw79m4wFp8KXh5lUfakgWw+28V3Ww1r0u2lU0DFGfNf6+YI:Ufj13Kgz+TK84i60B3D1l0w7cER2djkqPwO/1349Dq+vmuffwQ5tFPG24CkqWwYk87GzshlKxf7NLAf58HXh8hF14QLA0Ck3QDwob3FW2AJA:5f3KX:lyG1:GfYogA+IHome3
Fv2jyVQwKd8xK3DgQ82Inkq8h8hIKfJGfJgCtL0lFvAlDu6Gn54vVaMa89k84fTAZTHDHPwXkTm5SgBn5cNL02ONWKK3W81Rge48wWagj0Kc2w7zt5b6tndPrk7a9C05+Wnj51be78D82hVJXWBE01Qhrw6reaNW/sG8R0wiftrBcbvutPjyFmSpAkng+FW0zHDJ
ZRL1H0Wgbytt7T8GrcqWgVWg8Cv8Nfg7lF72G10G02c0u8HtXy158G0apPhQcpjwGphVndLcikiDptdrfCw== ubuntu@ip-172-31-17-136
ubuntu@ip-172-31-17-184:~/.ssh$
```

## Test SSH Connection

Step 5: Run command : `ssh ubuntu@instance2_private_ip`

With this we should be able to connect to instance 2 from instance 1

```
ubuntu@ip-172-31-17-136:~$ ssh ubuntu@172.31.17.184
The authenticity of host '172.31.17.184 (172.31.17.184)' can't be established.
ED25519 key fingerprint is: SHA256:lhUTS2dNKEG90B+KQWgCiCS/sMnahgMSFk412zta3E
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.17.184' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 14 12:43:42 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.7% of 6.61GB Processes:    121
Memory usage: 27%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.17.184

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Fri Aug 14 12:39:46 2026 from 18.206.107.27
ubuntu@ip-172-31-17-184:~$
```

i-080346908ec538cfe (ubuntu\_server\_1)  
PublicIPs: 3.90.252.110 PrivateIPs: 172.31.17.136

Step 6: Create a File on ubuntu Instance-1

`echo "Hello from Instance1" > file2.txt`

step 7: Copy File Using SCP

Run command : `scp file2.txt ubuntu@<Instance2-Private-IP>`

```
ubuntu@ip-172-31-17-184:~$ echo "I am Sujata from instance 1" > file2.txt
ubuntu@ip-172-31-17-184:~$ scp file2.txt ubuntu@172.31.17.184
ubuntu@ip-172-31-17-184:~$
```

i-080346908ec538cfe (ubuntu\_server\_1)

PublicIPs: 3.90.252.110 PrivateIPs: 172.31.17.136

Step 8: Verify on Instance-2

Run command :

`ll -l`

`cat file.txt`

output should be : Hello from Instance1

```
ubuntu@ip-172-31-17-184:~/.ssh$ cd ..
ubuntu@ip-172-31-17-184:~$ ll
total 48
drwxr-x--- 4 ubuntu ubuntu 4096 Aug 14 12:56 ./
drwxr-xr-x 3 root    root   4096 Aug 14 12:34 ../
-rw----- 1 ubuntu ubuntu   41 Aug 14 12:38 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Aug 14 12:35 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Aug 14 12:42 .ssh/
-rw----- 1 ubuntu ubuntu 1220 Aug 14 12:42 .viminfo
-rw-rw-r-- 1 ubuntu ubuntu   19 Aug 14 12:46 file.txt
-rw-rw-r-- 1 ubuntu ubuntu   28 Aug 14 12:56 file2.txt
-rw-rw-r-- 1 ubuntu ubuntu   28 Aug 14 12:57 ubuntu@172.31.17.184
ubuntu@ip-172-31-17-184:~$ cat file2.txt
I am Sujata from instance 1
ubuntu@ip-172-31-17-184:~$
```

|===== L5 completed =====|

## L6 - Write a Linux Shell Script to Install Git, JDK, Maven in EC2 Ubuntu Instance

Step 1: Create a shell script

```
vim install.sh
```

Step 2: Identify the commands to install the softwares

```
sudo apt install git -y
```

```
sudo apt install openjdk-25-jre-headless -y
```

```
sudo apt install maven
```

Step 3: Write a script

```
#!/bin/bash

#Exit script if any command fails
set -e

echo "Update Ubuntu packages"

sudo apt update -y

echo "======"

echo "Install git"

sudo apt install git -y

echo "Install Java"

sudo apt install openjdk-25-jre-headless -y

echo "Install Maven"

sudo apt install maven

echo "======"

echo "======"

echo "Verify Installation"

echo "======"
```

```
git --version
```

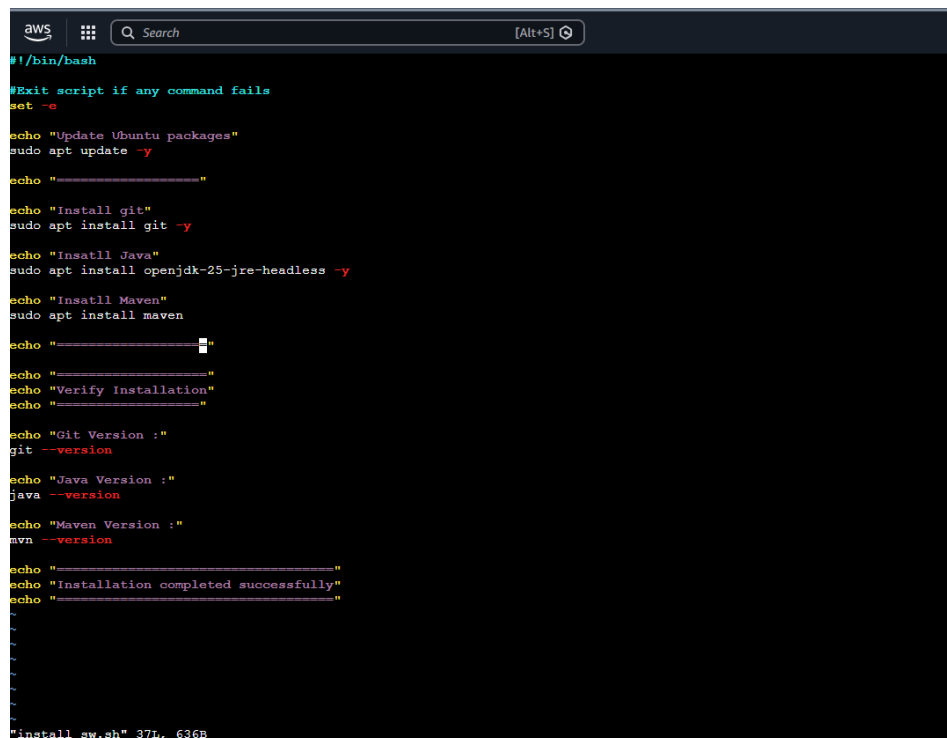
```
java --version
```

```
mvn --version
```

```
echo "====="
```

```
echo "Installation completed successfully"
```

```
echo "====="
```



Result :

```
ubuntu@ip-172-31-29-160:~$ cd script/
ubuntu@ip-172-31-29-160:~/script$ sh install_sw.sh
Update Ubuntu packages
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
117 packages can be upgraded. Run 'apt list --upgradable' to see them.
=====
Install git
git is already the newest version (1:2.53.0-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 117
Install Java
openjdk-25-jre-headless is already the newest version (25.0.3+9-2~26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 117
Install Maven
maven is already the newest version (3.9.12-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 117
=====
Verify Installation
=====
Git Version :
git version 2.53.0
Java Version :
openjdk 25.0.3 2026-04-21
OpenJDK Runtime Environment (build 25.0.3+9-2-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 25.0.3+9-2-26.04.2-Ubuntu, mixed mode, sharing)
Maven Version :
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 25.0.3, vendor: Ubuntu, runtime: /usr/lib/jvm/java-25-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-1006-aws", arch: "amd64", family: "unix"
=====
Installation completed successfully
=====
ubuntu@ip-172-31-29-160:~/script$
```

|===== L6 completed =====|